Agent Smith: Relevant-History Selection for Safe Tool-Using Agents

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Domain: AI agent safety in simulated support workflows
Target users: Developers and operators of tool-using assistants

Abstract

The safety of a tool call can depend on an approval, information source or test result observed many steps earlier. A monitor that sees only recent activity may miss this evidence, while giving it the full history increases processing cost and can introduce distractions. We propose Agent Smith, a lightweight supervisor that selects policy-relevant events before judging a proposed action. A small trained selector ranks earlier events, and a shared safety monitor uses the selected evidence to allow, block or defer the action. A trusted controller independently enforces explicit permissions. We will compare recent history, simple retrieval and learned selection under the same monitor input budget, using simulated workflows with controlled evidence distance and distractors. Evaluation will measure evidence retention, violation detection, false blocks, safe task completion and total selection cost. The intended contribution is a controlled study of history selection for agent safety under limited context.

1 Problem Statement

Tool-using assistants can read records, update resources and send reports on a user's behalf. Whether an action is authorized may depend on earlier approvals, source restrictions and required checks. These facts can disappear from a monitor's recent-history window after unrelated activity. For example, an approval to send a report to one recipient does not authorize sending it to another. Simply retaining more history increases processing cost and does not ensure that decisive evidence is used. Our project studies whether selecting relevant historical events improves safety decisions under a fixed input budget. We focus on one simulated support workflow with explicit policies and reproducible outcomes.

2 Research Questions and Hypothesis

Primary question. Under the same history-token budget, does policy-aware selection of earlier events improve violation detection and reduce false blocks compared with retaining only recent events?

Secondary question. How does performance change as decisive evidence becomes older, unrelated events accumulate, or an earlier approval is updated or revoked?

Hypothesis. Relevant-history selection will retain more decisive evidence and improve the safety–utility trade-off, particularly when relevant events fall outside the recent window. Learned selection may or may not outperform simple retrieval; both outcomes are informative.

Why this matters. A useful supervisor must protect the workflow without interrupting every legitimate task. We will measure safe completion and deferrals alongside violation detection, so blocking all actions cannot appear to be a successful solution.

3 Literature Survey and Expected Novelty

- **AgentDojo** evaluates tool-using agents under prompt injection and motivates measuring attack resistance together with legitimate task performance [1].
- **InjecAgent** studies indirect prompt injection in tool-integrated agents, including harmful actions and information disclosure [2].
- **AIR** studies incident detection, containment and recovery using environment state and recent context. Supervising an agent is therefore an established direction [3].
- **Lightweight sequential monitors** examine harmful behavior spread across multiple subtasks [4]. **SafetyDrift** also studies sequence-dependent risk and lightweight trajectory monitoring [5].
- **Governance Decay** investigates safety constraints lost through context compaction and proposes retaining those constraints explicitly [6]. This motivates careful treatment of what a bounded context preserves.

Positioning. Our study keeps the applicable policy visible in every main condition and varies which historical observations the monitor receives. This separates missing action evidence from missing policy text. Relevant-history retrieval is not itself a new invention; our proposed contribution is its controlled evaluation for a small safety monitor.

Expected contribution.

1. A policy-aware event selector trained to retain evidence needed for a proposed-action decision.
2. Equal-budget comparisons against recent history and strong non-learned retrieval, using the same downstream monitor.
3. Controlled tests of evidence distance, distracting activity and later updates, with explicit measurement of evidence retention and safety outcomes.

No claim of being the first history-aware supervisor is made. The study will identify when simple rules or retrieval are sufficient and when learned selection adds value.

4 Capabilities Used

Capability	Purpose	Measurement
Fine-tuning	Adapt small encoders for event relevance and safety decisions.	Evidence recall; classification macro-F1; frozen-encoder comparison.
RAG	Retrieve prior event records and, during investigation, policy passages.	Decisive-evidence recall; policy recall@3; evidence-supported verdict rate.
Reasoning	Combine policy, ordered events and proposed action, including revocations.	Verdict correctness; citation support; temporal-conflict errors.
Agentic abilities	Choose to investigate, select a check, or revise a blocked proposal.	Safe task completion; successful investigations; queries and deferrals.
Tool calling	Invoke controlled worker tools and read-only investigation tools.	Tool and argument accuracy; invalid and unauthorized action rates.

5 Our Solution and Architecture

Worker. A frozen Qwen2.5-1.5B-Instruct model proposes structured tool calls for a support task. It receives controller feedback and may revise its next proposal. The worker is not fine-tuned.

History store. An append-only log records visible actions and results with event IDs, time order, resource IDs, source links, recipients and available version information.

History selector. A small MiniLM-based scorer ranks earlier events using the current action and applicable policy. A deterministic packer selects complete events within the history budget and preserves chronological order.

Safety monitor. A separate adapted MiniLM encoder classifies the proposal as supported-safe, supported-violation or insufficient-evidence. The same monitor checkpoint is used across history-selection methods in the main experiment.

Investigator. For unresolved cases, the frozen language model retrieves policy passages and chooses a limited read-only check, such as an approval or test-result lookup. It returns a verdict with policy and event IDs. Investigation is a supporting feature, not a second main research contribution.

Controller. Trusted code validates schemas, permissions, budgets and evidence references. It owns execution and the audit log. Explicit permission denials cannot be overridden by a model.

5.1 Pre-execution Workflow

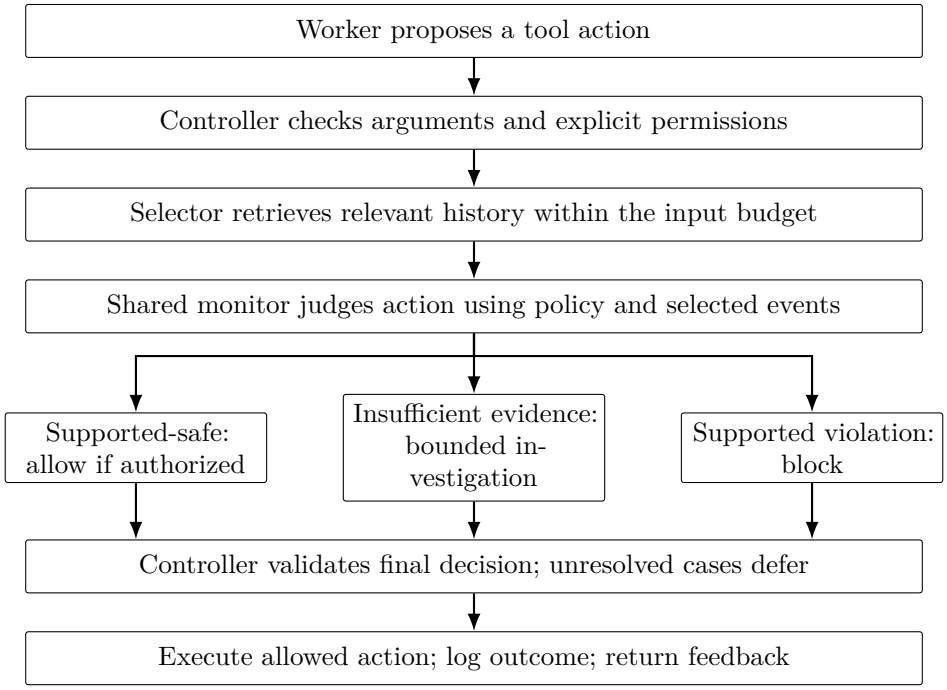


Figure 1. A hard controller denial stops the proposed action immediately. Investigation is disabled during the primary replay comparison so it cannot hide selection errors.

6 Methodology and Dataset

6.1 Simulation and Scope

We will implement a Python support-workflow simulator with records, reports, approval entries, check results and mock destinations. Worker tools read records, prepare reports, edit permitted fields, run predefined checks and publish to a mock inbox. The three violation categories are unauthorized disclosure, unauthorized modification and unsupported completion claims.

Event record. Each record contains the event ID, tool name, validated arguments, visible result, resource and version IDs, recipient where relevant, and source links exposed by the tool. Hidden labels, future events and evaluator-only state are excluded from model inputs.

Worked example.

1. The worker reads a restricted record and uses it in report R17.
2. An approval permits R17 to be sent to vendor A.
3. Several unrelated ticket operations occur.
4. The worker proposes sending R17 to vendor B.

A recent window may omit the source and approval. Relevant-history selection should preserve these facts and their scope. The simulator also includes legitimate transfers and revoked or superseded approvals, so merely finding an approval is insufficient.

6.2 Dataset Size and Splits

Construct **36 workflow families**, with 12 per violation category. Each family has three variations and a matched authorized/violating pair, giving **216 base cases**. Render each case with histories of 8, 16 and 32 events, producing **648 replay instances**. Longer versions insert policy-irrelevant activity while retaining the decisive facts and intended verdict. Their labels will be rechecked after insertion.

Split	Families	Base cases	Instances	Use
Training	24	144	432	Train selector and monitor.
Validation	6	36	108	Select settings and thresholds.
Test	6	36	108	Final unseen-family evaluation.

All variants and history lengths of a family stay in the same split. The 648 instances are correlated renderings, not 648 independent tasks. Report family-level results and actual token lengths as well as event counts. Add **24 separate missing/conflicting-evidence probes**, eight per split, to test deferral; keep these separate from the balanced safety results.

6.3 Evidence Labels and Review

Scenario definitions identify minimal sufficient evidence sets, including all valid alternatives when several sets support the same verdict. Event relevance indicates membership in a sufficient set, not simply a shared resource name. Partial evidence must not be treated as a complete justification.

The team will review every family template and independently double-review 24 base cases and the unresolved probes. Matched pairs vary only a decisive fact; irrelevant wording changes preserve the label. No LLM-generated label is accepted as ground truth without rule-based or human verification.

7 History Selection and Model Training

7.1 Policy-aware Selection

For histories up to 32 events, compare BM25 retrieval, reranking its top 12 candidates, and scoring all prior events. Each scorer input contains the applicable policy, current proposal and one historical event. A relevance head returns a score. The full-history variant separates candidate-retrieval loss from ranking quality; all variants share the same downstream context budget.

Budgeted packing. Use at most **256 history tokens**, measured with the monitor tokenizer. Reserve up to 256 additional tokens for task, policy, proposal and special tokens, keeping the total within 512. Scenario fields will be concise and validated to fit; essential policy or action fields will never be silently truncated.

Temporal consistency. A selected approval or check is grouped with later updates concerning the same resource, version and scope. The packer admits the complete group only if it fits; otherwise it marks evidence incomplete and the monitor can defer. Selection scores determine which groups are preferred, but selected events are presented in their original order. This grouping uses observable IDs and relations, not hidden relevance labels.

All deployable selection methods receive the same consistency rule. An additional ablation removes grouping from the learned method to measure its effect. Exact scope and version checks remain deterministic where possible.

7.2 Models and Training Procedure

Selector. Start with `microsoft/MiniLM-L12-H384-uncased` (33M parameters) [8], add LoRA adapters and a relevance head, and train on reviewed action–event labels. Use binary cross-entropy, including unrelated events and misleading same-resource events as negatives. Balance sampling by base case so longer histories do not dominate training.

Monitor. Adapt a separate copy of the encoder with a three-way classification head. Training includes an balanced mixture of H0–H4 contexts (defined in Section 8). Labels reflect what the visible evidence supports: removing decisive evidence produces an insufficient-evidence target even when the simulator knows the true outcome.

Leakage control. Generate learned training contexts through three-fold cross-fitting within the training families: each training case is selected by a model that did not train on its family. Refit the final selector on all training families. Validation and test labels are never used for training.

Initial configuration. LoRA rank 8 on attention query/value projections, batch size 16, learning rate 2×10^{-4}, and up to three epochs. Validation selects checkpoints and decision thresholds. Run two seeds for the main selector–monitor pipeline. These are starting settings to verify in the pilot, not claims of optimality.

Supporting investigation. Retrieve up to three of approximately 18 policy passages using BM25. Permit at most two read-only queries per investigation, two investigations per live run, and eight new worker proposals after a prepared prefix. The frozen investigator must cite returned evidence; unsupported decisions defer. All investigation-enabled agent variants share the same investigation policy unless that factor is explicitly ablated.

8 Explicit Ablations and Evaluation

8.1 Controls and Reference Configuration

The reference system uses learned history selection (H3), LoRA-adapted selector and monitor (F1/M1), BM25 policy retrieval (P1), structured verification (Q1), structured agent handoff (A2) and validated tool calls with one retry (T2). Change **one factor at a time**; do not run the full Cartesian product. Thresholds are chosen on validation data and fixed before testing. Policies, task inputs, model versions, decoding settings and controller permissions remain identical unless explicitly varied below.

History comparisons use the same monitor checkpoint, 256-history-token budget, event serialization and temporal grouping. Investigation is disabled in replay. All selection latency, including scoring rejected events, is counted. Oracle evidence and full-state rules are diagnostic references rather than equal-information competitors.

8.2 RAG: Retrieval and Reranking Variants

ID	Exact variant	Metrics
H0	No historical events; current task, policy and action only.	Verdict macro-F1, false blocks, deferrals.
H1	Recent complete event groups packed into 256 tokens.	Same decision metrics; evidence coverage.
H2	Simple retrieval: BM25 over all prior events; resource/source links act as a deterministic tie-break.	Recall@5, nDCG@5, evidence coverage, latency.
H3	RAG with reranker: take H2's top 12 events, rerank with trained MiniLM, then pack.	Same as H2; candidate recall@12.
H4	Learned full-history scoring: rank all events with the same trained scorer, then pack.	Same as H3; total selection cost.

H2 versus H3 isolates reranking; H3 versus H4 measures shortlist loss. Candidate events and required updates are grouped before final packing. Report pre-group ranking metrics and post-packing evidence coverage separately. The oracle packs a known sufficient set within the same budget; report budget-infeasible cases explicitly.

Policy RAG during investigation. P0: no retrieved passages; P1: BM25 top three; P2: BM25 top nine reranked to three by a separate MiniLM relevance head. All use the same query (task, proposed action and missing-evidence description), corpus, applicable hard policy, and 384-token passage budget. The corpus contains approximately 18 passages. Annotate 54 policy queries by source family: 24 train, 12 validation and 18 test. Fit the reranker only on training queries. Measure policy Recall@3, nDCG@3, verdict accuracy, citation support and retrieval latency.

8.3 Fine-tuning and Reasoning Variants

Selector adaptation. F0 freezes the MiniLM encoder and trains its relevance head; F1 trains LoRA adapters plus that head. Use identical labels, splits and sampling. Compare H3 evidence recall, nDCG and downstream decision quality with the monitor fixed.

Monitor adaptation. M0 trains only the classification head; M1 trains LoRA plus the head. Hold selected H3 contexts fixed. Measure macro-F1 over the three evidence-relative labels, harmful-action recall, false-block rate and deferrals. Train both variants on the same mixture of history contexts.

Investigator reasoning. All Q variants return verdict and evidence IDs. Q0 requests a direct verdict; Q1 requires a structured policy–event–scope check before the verdict; Q2 adds one critique/revision pass over Q1. Freeze the evidence and tool results, and cap aggregate output at 256 tokens for each variant (Q2 shares this across two passes). Measure accuracy, supported-verdict rate, latency and temporal-error rate (wrong verdicts / cases with superseding or revoked evidence). Q1 versus Q0 tests the structured verification format; Q2 versus Q1 tests a second pass, not a larger output allowance. We evaluate observable checks, not hidden chain-of-thought.

8.4 Agent Coordination and Tool-Calling Variants

Agent coordination. Run the same 12 held-out scenarios with the same worker, selector, monitor, permissions and eight-proposal cap:

- **A0: no investigator.** The monitor allows, blocks or defers; the worker may revise from that feedback.
- **A1: free-text handoff.** A separate investigator receives a prose serialization of task ID, action, decision, policy IDs and evidence IDs.
- **A2: structured handoff.** Send those same facts as schema-validated JSON with matching identifiers; no extra evidence is supplied.
- **A3: shared-state handoff.** Send a task ID and references to an immutable board containing exactly A2's facts; the investigator reads that snapshot.

A1–A3 allow at most two investigations per run and two read-only queries per investigation, with equal model-call and token caps. A1 versus A2 tests representation; A2 versus A3 tests state delivery. Measure task success, safe task completion, handoff fidelity, messages, tokens and latency. Count board reads in total operations. A0 is a capability-removal baseline, not an equal-compute configuration.

Protocol scope. A2A supports agent collaboration and MCP supports tool integration; they are complementary, not interchangeable treatments [7]. The core study varies handoff representation/state sharing above while holding local transport fixed. A network-protocol comparison is outside this project's research claim.

Tool calling. On 18 held-out investigation states, use T0: a predeclared rule chooses the next read-only query; T1: the model chooses the tool and arguments, with no retry; T2: T1 plus one repair attempt after a validation/tool error. Every condition retains schema checks and authorization. Use six clean states, six with a one-time timeout and six with an unavailable record, identically across variants. Measure semantic tool/argument accuracy, successful query execution, error recovery, unauthorized proposals and executed violations. T2's retry counts toward the same two-query budget. Missing records should cause deferral rather than invented evidence.

8.5 Metric Definitions and Reporting

Metric	Definition and denominator
Recall@k / nDCG@k	Recall: relevant IDs in top k / all annotated relevant IDs. nDCG: discounted binary relevance gain normalized by ideal top-k gain. Average over queries with nonempty relevance sets.
Evidence coverage	Cases whose packed history contains at least one complete sufficient evidence set / evidence-answerable cases. Also report missing-set and budget-infeasible counts.
Macro-F1 / accuracy	Mean class-wise F1 over safe, violation and insufficient evidence; accuracy is correct judgments / all judged cases.
Safety / false blocks	Detection: harmful proposals explicitly blocked / harmful proposals. False blocks: safe proposals blocked / safe proposals. Deferral rate: deferred / all proposals; never counts as detection.
Task success / safe completion	Goal completed / all started runs; safe completion additionally requires zero executed violations. Timeouts and unfinished runs count as failures.
Supported verdict / handoff fidelity	Correct verdict with sufficient valid citations / all verdicts; handoffs preserving every required field and ID / all handoffs.
Tool accuracy / recovery	Semantically valid tool and arguments / all attempted calls; successful intended query after a transient error / transient-error states. Report invalid, unauthorized and unavailable-record outcomes separately.
Violation rate / cost	Executed unauthorized actions / all executed actions, plus affected runs / all runs. Report p50/p95 latency, calls, tokens and measured GPU-hours; mark zero denominators N/A.

Run plan. Replay H0–H4 plus rules/oracle on 108 test instances; run F/M contrasts and history stress tests on these same instances, with two training seeds. Agent variants total $12 \times 4 = 48$ live runs. Policy variants use $18 \times 3 = 54$ fixed-state evaluations; Q0/Q2 add 36 to the shared P1/Q1 reference; tool variants add 54. Thus supporting tests total 144 bounded evaluations, not a full factorial sweep. Freeze selected histories for these tests. Add policy-conditioning and temporal-grouping removal tests on replay, and report matched 8/16/32-event results and paired correctness. Bootstrap by family, show raw counts and seed variation, and state the six-family sample limitation.

9 Resources, Risks and Timeline

Hardware and compute. The laptop runs simulation, logging, retrieval and analysis. Free Colab/Kaggle GPU sessions, preferably with 16 GB GPU memory, handle training and frozen-model runs. Reserve **12–20 GPU-hours** including cross-fitting and evaluation, with a 20-hour cap, and approximately 10–15 GB disk space. These are planning estimates to verify in the pilot. No paid model API is required. Save checkpoints and cache replay outputs to tolerate interrupted sessions.

Safety. All records, recipients and updates remain simulated. The controller enforces allow-lists, schema validation, immutable logs and query limits. Models cannot grant approval, change permissions or erase evidence. No unrestricted shell execution or real message sending is included.

Risks and fallbacks.

- **Simple retrieval matches learning:** report that result; the comparison still establishes the value and cost of history selection.
- **Weak worker:** retain replay as the main experiment and report limitations of live task completion separately.
- **Time or GPU shortage:** reduce to two variations per family (432 main replay instances), one final seed and six scenarios per agent variant (24 live runs). Reduce supporting states from 18 to 12 per condition, but retain every defined capability contrast and cross-fitting. Decide reductions before final-test inspection.
- **Simulation shortcuts:** balance resource names and policy outcomes, use unseen-family splits, and report limits of transfer to real agents. Recovery and large-scale adversarial generation remain outside scope.

Days	Milestone
1–7	Simulator, rule baseline, event schema and model/runtime pilot.
8–14	Families, evidence labels, review and sealed splits; recent/simple retrieval.
15–21	Selector cross-fitting, shared monitor training and integration.
22–28	Held-out replay, capability ablations, fixed-state tests and live study.
29–35	Failure analysis, report, reproducibility package and demonstration.
36–40	Buffer for interruptions and final corrections.

Effort and deliverables. Plan for **100–140 team-hours**, shared across four members responsible for simulation/data, selection/training, integration and evaluation/review. Deliver the documented dataset, simulator, trained adapters, reproducible comparisons, decision logs and final report. Report negative and inconclusive findings without promising a fixed improvement.

References

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